

Chapter 3

Networked Communications



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3.1 Introduction

ON THE EVENING OF SEPTEMBER 6, 2022, THE CELLPHONES OF 27 million californians simultaneously blared an alarm. It wasn't an Amber Alert, though. Instead, the text read: "Power interruptions may occur unless you take action." An extreme heat wave had increased power consumption in the state to 52,000 megawatts, and the electrical grid was maxed out. Fortunately, the emergency text message worked where ordinary appeals had failed. Within 45 minutes, residents reduced their power consumption by 2,600 megawatts, and rolling blackouts were averted [1].

M- Pesa, Africa's first mobile money platform, allows people to conduct financial transactions using their mobile phones. Millions of Africans who have never had a bank account use their cell phones to save money, pay bills, and transfer funds over large distances. In Kenya, where nearly every household has an M-PESA account, access to mobile money has reduced poverty by encouraging saving and supporting entrepreneurship. Nearly 200,000 Kenyan women have used

M-PESA to raise their families' incomes by shifting from subsistence farming to business or sales occupations [2].

In 2017 the Royal Society for Mental Health interviewed 14–24-year-olds in the United Kingdom to better understand how their use of social media affected their mental health and sense of well being. The survey revealed that 91 percent of young people used social media sites. The report noted some positive effects from the use of social media: allowing young people to grow real-world relationships; serving as a vehicle for positive self-expression; providing the opportunity to learn about other people's mental health issues and experiences; and delivering emotional support during difficult times. However, the report also warned that the heavy use of social media can have detrimental effects on the mental health of young people, by increasing feelings of anxiety, lowering the quality of sleep, raising concerns about body image, exposing them to cyberbullying, and heightening the fear of missing out [3].

Before the 2016 presidential election, more than 100 pro-Trump Web sites were being managed from Veles, a small town in Macedonia. The Macedonians were not Trump supporters, but they found the creation of these sites to be an easy way to make money. They plagiarized stories from right-

wing sites in the United States, added sensationalized headlines, posted the stories to their sites, and then used the “share” feature of Facebook to generate traffic to the stories. When people clicked on their stories, Google paid them for the ads placed on their pages [4].

5 Watching other people compete at video games, or **eSports**, began with the Internet café culture in South Korea and China and has now become a global phenomenon. Professional eSports leagues are growing in popularity, and companies throughout North America are transforming movie theaters, parking garages, and other large spaces into eSports arenas. Twitch is the most popular platform for live streaming video games over the Internet. The most popularly watched eSports event in 2022 was the League of Legends World Championship, which attracted more than 5 million viewers [5].

These stories illustrate how the Internet and cellular networks are changing life in the modern world ([Figure 3.1](#)). The development of these networks has provided us with exciting new ways to conduct business, connect with friends, and entertain ourselves. However, other fruits of the same technologies are bitter: new ways of scamming people, new

ways to bully and harass others, and new concerns about the impact of social media on democratic institutions.

Figure 3.1



Far more people in the world have access to cellphones than to electricity or clean water.

(Charles Sturge/Alamy Stock Photo)

In this chapter we explore how people use and abuse the Internet and cellular networks. We discuss how people communicate with one another and how they organize and find information. Some governments are threatened by the way in which modern networks have supported these activities; we detail various kinds of censorship and consider appropriate limits to the freedom of expression. We also discuss and evaluate new behaviors that have emerged with the growth of these networks, such as sexting, cyberbullying, and Internet addiction.

3.2 Spam

Over 4 billion people around the world use email, and every day about 350 billion email messages are sent [6].

Unfortunately, a significant percentage of this traffic consists of unsolicited bulk email, or **spam**.

Why is spam called spam? Brad Templeton, chairman of the board of the Electronic Frontier Foundation, traces the term back to the SPAM sketch from *The Final Rip Off* by Monty Python's Flying Circus, in which a group of Vikings drown out a café conversation by loudly and obnoxiously repeating the word “spam” [7]. In a similar way, legitimate email messages can get “drowned out” by spam.

3.2.1 The Spam Tsunami

The rise of spam corresponded with the transformation of the Internet from a noncommercial academic and research enterprise into a commercial global network. Early spam messages provoked Internet users and generated big headlines. For example, in 1994 Phoenix lawyers Laurence Canter and Martha Siegel sent an email advertising their immigration services to more than 9,000 electronic

newsgroups. Canter and Siegel received tens of thousands of responses from outraged newsgroup users who did not appreciate seeing an off-topic, commercial message. The *New York Times* reported the incident with the tongue-in-cheek headline, “An Ad (Gasp!) in Cyberspace.” Canter and Siegel were undeterred. Their ad was successful in bringing them new clients. “We will definitely advertise on the Web again,” Canter said. “I’m sure other businesses will be advertising on the network in the very near future” [8]. Still, spam was not seen as much of a problem throughout the 1990s, and even in the year 2002 it accounted for only about 9 percent of all email traffic [9].

The spam tsunami hit in 2003, with a big jump in the volume of spam messages, and by 2009 about 85 percent of all email messages were spam [10]. Spam consumed a large percentage of the Internet’s bandwidth and huge amounts of storage space on mail servers and individual computers.

To deal with the deluge, an entire industry grew up developing software designed to filter out spam. These filters looked for a large number of messages coming from the same email address, messages with suspicious subject lines, or messages with spamlike content. ISPs and businesses installed spam filters to keep spam from reaching users’ mailboxes.

Even with spam filters in place, the cost to businesses in wasted productivity from the spam that got through was estimated at tens of billions of dollars per year in the United States alone [11].

The volume of spam grew so large because spam is effective. The principal advantage of spam is its low cost compared to other forms of advertising. For about \$10, a company can send an advertisement to a million different email addresses [12]. Sending the same advertisement to a million addresses using the US Postal Service costs at least \$40,000 for the mailing list and \$280,000 for bulk-rate postage. And that doesn't include the cost of the brochures! In other words, an email advertisement is more than 30,000 times cheaper than a traditional flyer sent out in the mail.

Where do spammers get email lists with millions of addresses? The Internet provides a variety of sources of email addresses that can be harvested and sold to spammers. For example, email addresses often appear in Web sites, in chat-room conversations, and newsgroups. Some computer viruses gather email addresses stored in the address books of PCs and transmit these addresses to spammers.

Another way to garner email addresses is through dictionary attacks (also called directory harvest attacks). Spammers

bombard Internet service providers (ISPs) with millions of emails containing made-up addresses, such as AdamA@isprovider.com, AdamB@isprovider.com, AdamC@isprovider.com, and so on. Of course, most of these emails will bounce back, because the addresses are no good. However, if an email doesn't bounce, the spammer knows there is a user with that email address and adds it to its mailing list.

Sometimes people voluntarily reveal their email address. Have you ever entered a contest on the Web? There is a good chance the fine print on the entry form said you agree to receive "occasional offers of products you might find valuable" from the company's marketing partners; in other words, spam [13]. Sign-ups for email lists often contain this fine print, too.

At the peak of the spam tsunami in 2009, about 90 percent of spam was being sent out by **botnets**: huge networks of compromised computers controlled by "bot herders." Bot herders create botnets by launching programs that search the Internet for computers with inadequate security and install software robot programs, called bots, on these vulnerable systems. A computer with the bot program installed on it is called a zombie because it can be directed by a remote

computer to perform certain tasks. Bot herders can send out billions of email messages every day by dividing the address lists among the hundreds of thousands of zombies they control [14].

For more than a decade, law enforcement agencies and Internet service providers around the world have engaged in a sustained effort to shut down botnet command centers [15]. In addition, the behavior of spammers appears to have changed after 2009, moving away from scattershot mailings to somewhat more selective email lists [15]. For both these reasons, the share of spam in email traffic peaked at around 85 percent in 2009 and has been trending downward ever since [10]. In 2022 spam accounted for about 49 percent of all email messages [16].

3.2.2 Need for Social-Technical Solutions

As we saw in [Chapter 1](#), new technologies sometimes cause new social situations to emerge. The spam epidemic is an example of this phenomenon. The Internet allows people to send email messages for virtually no cost. Because a spammer's profits increase as the number of sent messages

increases, every spammer has an incentive to send as many messages as possible.

The spam problem arose because the Internet and email technology developed without taking social expectations into account. The design of the Internet allows sophisticated users to disguise their own email addresses. Spammers take advantage of this loophole to send out millions of messages, knowing that unhappy recipients will not be able to respond. This is contrary to a fundamental social expectation: fairness. In order to be fair, communications should be two-way, not one-way [17].

3.2.3 Case Study: Anh the Acme Accountant

Anh is an accountant at Acme Corporation, a medium-sized firm with 50 employees. All the employees work in the same building, and Anh knows all of them on a first-name basis. In fact, Anh distributes paychecks to Acme's employees at the end of every month.

Anh's 10-year-old daughter is a Girl Scout. During the annual Girl Scout cookie sale, Anh sent an email to all the other Acme employees, inviting them to stop by her desk

during a break and place orders. (There is no company rule prohibiting the use of the email system for personal emails.) Nine of the recipients were happy to get Anh's email, and they ordered an average of four boxes of cookies, but the other 40 recipients did not appreciate having to take the time to read and delete an unwanted message; half of them complained to a coworker about Anh's action.

Did Anh do anything wrong?

Kantian Analysis

According to the second formulation of the Categorical Imperative, we should always respect the autonomy of other people, treating them as ends in themselves and never only as the means to an end. The story provides evidence that Anh was not simply "using" her coworkers as the means to her end of making money for the Girl Scouts. She didn't misrepresent what she was doing. She didn't force anyone to buy the cookies or even read the entire email; employees not interested in Girl Scout cookies could simply delete Anh's message as soon as they read the subject line. Some people who received the email freely chose to buy some cookies. Therefore, what Anh did wasn't strictly wrong.

On the other hand, if Anh had found a way for those people interested in hearing about the Girl Scout cookie drive to “opt in” to her announcement, those people not interested in purchasing Girl Scout cookies would not have been bothered by her email. An “opt in” approach would have been better because it would have shown more respect for the time of Anh’s coworkers.

Act-Utilitarian Analysis

We will do our evaluation in terms of dollars and cents, quantifying the benefits and costs of Anh’s action. Let’s begin with the benefits. A box of cookies costs \$4 and provides \$3 of profit to the Girl Scouts. Someone who buys a box of Girl Scout cookies understands it is a fund-raising activity and is happy with what they receive for \$4. Since the cost of \$4 is matched with \$4 of benefit, they cancel each other out in our analysis, and we do not have to worry about this factor anymore. The average employee who participated in the sale purchased four boxes of cookies. Nine employees participated, which means Anh sold 36 boxes of cookies and provided \$108 of benefit to the Girl Scouts.

Now let’s look at the harms. The principal harm is going to be the time wasted by Acme’s employees. Anh took orders

and made deliveries during coffee or lunch breaks, rather than on company time, so our focus is on the 40 employees who did not appreciate getting Anh's solicitation. It's reasonable to assume that they spent an average of 15 seconds reading and deleting the message. That adds up to 10 minutes of lost productivity.

Half of the employees spent 5 minutes complaining about what Anh did with a coworker. You can imagine the typical conversation. "What makes her so special?" "How does she get away with this kind of thing?" "If I did this for my kid, I'd get in trouble." Taking both the employee's time and the coworker's time into account, Acme loses 10 minutes of productivity for each conversation. Multiplying 10 minutes by 20 conversations gives us 200 minutes.

) The total time wasted equals 210 minutes or 3.5 hours. Assume the average Acme employee makes \$20 per hour. The cost of the lost productivity is 3.5 hours times \$20 per hour or \$70.

The benefit of \$108 exceeds the cost of \$70, so we may conclude that Anh's action was good. We should note, however, that all the benefit went to the Girls Scouts and all the cost was borne by Acme Corporation. It would be perfectly reasonable if the owners of Acme Corporation

concluded that this kind of activity was not in the best interests of the company and created a new policy forbidding the use of company email for cookie drives and other fund-raisers.

Rule-Utilitarian Analysis

What would the consequences be if everyone used the company email system to solicit donations to their favorite causes? All the employees would receive many more messages unrelated to business. There would be plenty of grumbling among employees, lowering morale. Reading and deleting these solicitations would waste people's time, a definite harm. It's unlikely that any one cause would do well if everyone was trying to raise money for their own charity. There is a good chance the owner would become aware of this problem, and a logical response would be to ban employees from sending out this kind of solicitation. Because the harms are much greater than the benefits, it is wrong to use the company email system to solicit donations to a charity.

Social-Contract-Theory Analysis

Acme Corporation does not have a prohibition against using the company's email system for personal business. You could say that by sending out her email solicitation, Anh was

exercising her right to free speech. Of course, she did it in a way that many people might find obnoxious, because even if they did not choose to read her entire message, they had to take the time to scan the subject line and delete it. Unlike spammers, however, Anh did not disguise her identity as the sender, thereby providing unhappy recipients with the opportunity to respond to her email and voice their disapproval of her solicitation. If many of the 40 people who did not appreciate receiving her email sent a reply communicating their displeasure, then Anh got a taste of her own medicine by having to wade through a bunch of unwanted email messages, and she may choose a better method of advertising the Girl Scout cookie drive next year. From a social-contract-theory point of view, Anh did nothing wrong.

Virtue-Ethics Analysis

Good coworkers exhibit many virtues, including honesty, dependability, fairness, friendliness, and respect for coworkers. Three that seem particularly important for this case study are honesty, fairness, and respect. Anh demonstrated honesty by being completely straightforward about the Girl Scout cookie drive with her fellow employees. However, it wasn't fair or respectful for Anh to use the email

system to promote her own daughter's fund-raiser when other employees have not used email for similar solicitations for their children. Anh clearly exercised poor judgment when she sent the email, since half of the company's employees felt the need to complain about it.

Looking at this scenario from a completely different angle, we consider Anh in her role as a parent. Good parents want what is best for their children, and Anh was undoubtedly thinking of her daughter's welfare when she decided to sell the Girl Scout cookies at her workplace. Perhaps her daughter has a quota to meet, or perhaps nice prizes are given to those who sell enough cookies. Viewed in this light, Anh's actions seem consistent with those of a loving parent. However, parents are also supposed to teach their children how to develop into capable and independent adults. Anh could have used the cookie sale as an opportunity to teach her daughter some of those lessons. After all, her daughter is 10 years old, certainly old enough to handle many of the tasks. Instead, Anh ran the entire cookie sale operation herself and simply handed her daughter the proceeds.

We conclude Anh demonstrated many, but not all, of the characteristics of a good coworker and a good parent in this episode. If Anh wanted to help her 10-year-old daughter sell

cookies, fine, but she should have found a way for her daughter to play a more active role in the cookie sale at Acme Corporation. For example, her daughter could have come in after school one day to deliver the cookies to the people who ordered them and collect their payments. In this way, Anh's daughter could have gained the satisfaction of knowing she had contributed a good portion of the time and effort needed to achieve the desired result. Furthermore, Anh should have found another way to advertise the sale that respected her workplace's culture of keeping the email system free from solicitations.

Summary

Although the analyses of Anh's action from the perspectives of these five ethical theories reached different conclusions, our analyses have shown she could have taken another course of action that would have been much less controversial. Since Anh has only 49 coworkers, it would not have been too difficult for her to find out who wanted to be notified the next time the Girl Scouts were selling cookies. She could have put a sign-up sheet on her desk or the company bulletin board, for example. By notifying only those people who signed up, Anh's emails would have been solicited and personal. She could still take advantage of the efficiency of

the email system without anyone objecting that she was “using” coworkers or contributing to lost productivity, meaning there would be much less chance of the company instituting a policy forbidding the use of its email system for fund-raising activities. Finally, Anh could have found a way to share the work with her daughter.

3.3 Internet Interactions

The Internet mediates communications and commerce among more than four billion people. In this section we review just a few of the myriad ways people are using the Internet to interact with others and gain access to information.

3.3.1 The World Wide Web

The creation of the World Wide Web stimulated a tremendous growth in the popularity of the Internet. Its creator, Tim Berners-Lee, initially proposed the Web as a documentation system for CERN, the Swiss research center for particle physics, but the creation of easy-to-use Web browsers made the Web accessible to “ordinary” computer users as well [18]. The Web is a hypertext system: a flexible database of information that allows Web pages to be linked to each other in arbitrary fashion. Web browsers such as Chrome, Edge, Safari, and Firefox allow people to traverse this hypertext system with ease.

Two attributes enabled the Web to become a global tool for information exchange. First, it is decentralized. An individual or organization can add new information to the Web without asking for permission from a central authority. Second, every object on the Web has a unique address. Any object can link to any other object by referencing its address. A Web object's address is called a **URL** (uniform resource locator).

3.3.2 Mobile Apps

People are spending more time on smartphones and tablets and less time on laptop or desktop computers. Using Web browsers on mobile devices can be awkward, and for this reason organizations produce **mobile apps**: software programs that are loaded onto mobile devices. Some mobile apps are stand-alone programs, but others connect to the Internet, allowing people to download and upload data. Mobile apps are a popular way to access the Internet because they can be optimized to make the best use of a mobile device's resources (limited screen size, touch interface, etc.) [19].

3.3.3 How We Use the

Internet

Intuitive Web browsers and mobile apps have made the Internet accessible to people with little or no formal computer training. Today billions of people access the Internet for a wide variety of purposes. Here are just a few examples of how people are using the Internet.

1. *We buy.*

Shopping sites enable us to view and order merchandise from the comfort of our homes. According to the US Census Bureau, ecommerce represented 15 percent of all retail sales in the United States in the first quarter of 2023 [20].

2. *We sell.*

Craigslist allows people to place free classified advertisements on the Internet. Begun by San Franciscan Craig Newmark in 1999, Craigslist is now available in about 90 countries and attracts about 50 billion page views a month [21].

3. *We socialize.*

The Internet has become a popular way for friends to keep in touch with each other. The most popular social

network is Facebook, with 3 billion monthly active users in January 2023 [22]. Another well-known social network is LinkedIn, which serves people looking for professional contacts.

In 2012 the Dutch airline KLM launched a program that allows ticketed passengers who have uploaded information from their Facebook or LinkedIn profiles to select seatmates based on the profiles provided by other passengers [23].

4. *We contribute content.*

Popular apps allow people to upload videos, photos, podcasts, or other digital content. Instagram, with 2 billion monthly users, allows its users to upload photos and videos and share them on social networking services, such as Facebook.

A **wiki** is a Web site that allows multiple people to contribute and edit its content. The most famous wiki is *Wikipedia*, an online encyclopedia. Relying on the submissions of hundreds of thousands of volunteers, *Wikipedia* has become by far the largest encyclopedia in the world. Seventy-six languages are represented by at least 100,000 articles, but by far the most popular language is English, with more than 6.6 million articles

written as of 2023. However, critics wonder about the quality of a reference work that allows anyone with a Web browser to contribute [24].

5. *We blog.*

A **blog** (short for “Web log”) is a personal journal or diary kept on the Web. Used as a verb, the word blog means to maintain such a journal. Blogs may contain plain text, images, audio clips, or video clips [25].

Some commentators use the term **Web 2.0** to refer to a change in the way people use the Web. Social networking services, wikis, Flickr, Reddit, and blogs illustrate that hundreds of millions of people are now using the Web not simply to access content but to build communities and upload and share content they have created.

6. *We visit secret Web sites.*

A **darknet** is a network of Web sites that can only be accessed through special software that encrypts messages and provides users with anonymity. Darknets are popular with those who have something to hide: people who engage in pedophilia, sell illegal drugs, traffic in illegal weapons, and the like. However, Edward Snowden’s disclosure of surveillance by the

National Security Agency has prompted some people to ponder whether everybody should be using darknets on the principle that the government should not be spying on law-abiding citizens [26].

7. *We engage in crowdsourcing.*

Crowdsourcing is an online method of getting goods or services from a large group of people.

The Waze app is an example of information crowdsourcing. People use Waze to find the quickest way to drive from one place to another. If they allow the app to run while they are driving, it sends their car's GPS coordinates to Waze, which computes vehicle speeds and then uses this information about traffic congestion to help other Waze users find the best route [27].

Financing a project through crowdsourcing is called **crowdfunding**. Kickstarter is an American corporation that supports crowdfunding. Since 2009, more than 22 million people have pledged over \$7 billion to successfully fund more than 242,000 projects in a wide variety of creative areas: music, film and video, games, publishing, art, design, technology, theater, comics, food, fashion, and more [28].

8. *We learn.*

When the COVID-19 pandemic struck, schools and universities shifted to online learning, and the popularity of Zoom skyrocketed.

9. *We explore our roots.*

In the past, genealogists interested in accessing US immigration and census records had the choice between mailing in their requests and waiting for them to be processed or visiting the National Archives and examining the documents by hand. Now FamilySearch.org offers free access to indexes to US census information from 1790 to 1950, allowing genealogical research to be performed remotely—and much more quickly—over the Internet.

10. *We enter virtual worlds.*

An **online game** is a game played on a computer network that supports the simultaneous participation of multiple players. The Internet enables people to play games with total strangers and makes it possible for aficionados of obscure games to find opponents. The most popular online games attract millions of players.

South Koreans have enjoyed eSports for decades. Cybercafés (called **PC bangs** in South Korea) have large-screen monitors enabling spectators to watch the game-play, which is full of virtual violence and mayhem. Some children spend up to 10 hours a day playing games, hoping to turn professional. Kim Hyun Soo, chairman of the Net Addiction Treatment Center, complains that “young people are losing their ability to relate to each other, except through games” [29]. We discuss the topic of Internet addiction in [Section 3.11](#).

The phenomenon of global online gaming has created a real economy based on virtual worlds. In 2005 the *New York Times* reported on Chinese “gold farmers” who worked 12 hours a day, 7 days a week, earning \$3,000 a year killing monsters, harvesting virtual gold coins and artifacts, creating powerful avatars, and selling them over the Internet [30]. More recently, many citizens of Venezuela, trying to survive in a highly inflationary economy, have turned to gold farming. A Venezuelan gold farmer playing Runescape can earn 12,000 Venezuelan bolivares (\$0.50) an hour, five times the minimum wage [31].

11. *We control the Internet of Things.*

An increasing number of non-IT devices—thermostats, appliances, lights, motion sensors, door locks, garage-door openers, and baby monitors, to name just a few examples—are being equipped with wireless connections to the Internet, forming an **Internet of Things**. These devices can be controlled from a Web browser, allowing people to oversee them even when they are not at home.

More significantly, some Internet-connected devices can be programmed to interact with each other without human intervention. Imagine devices at home that can monitor the GPS coordinates of the owner's smartphone. An individual could drive home after a long, hot day at work. When they are 15 minutes away from home, the air conditioning in their home turns on. They turn onto their street, and their outdoor light turns on. As they pull into their driveway, the garage door opens automatically [32].

12. *We pay our taxes.*

About 94 percent of Americans' federal income tax returns for tax year 2021 were filed online [33].

13. *We gamble.*

Internet gambling is a \$94 billion-a-year global business [34].

14. *We take humanitarian action.*

Kiva is a Web site supporting person-to-person microlending. Kiva works with microfinance institutions to identify entrepreneurs from communities with relatively little wealth, and it posts information about these entrepreneurs on its Web site. People who wish to make an interest-free loan are able to identify the particular person to whom they would like to lend money. Lenders have the ability to communicate with the entrepreneurs and see the impact their loans are having on the recipients, their families, and their communities [35].

3.4

Cryptocurrencies

3.4.1 Features of Cryptocurrencies

The Internet has supported the development of **cryptocurrencies**: virtual currencies independent of any bank or government. Bitcoin, released in 2009, was the first cryptocurrency. There are now more than 19,000 cryptocurrencies, and more than 40 of them had market capitalizations of at least \$1 billion in 2022, although Bitcoin dominates the other cryptocurrencies. Its market capitalization is nearly as large as that of all the other cryptocurrencies put together [36].

The primary features of a cryptocurrency are:

- *Global Accessibility*: Anyone with an Internet connection can send and receive cryptocurrency.
- *Decentralization*: All transactions are recorded in an encrypted public ledger distributed across many

Internet-connected computers.

- *Anonymity*: Cryptocurrency transactions are recorded using a digital address rather than a personal identifier.
- *Security*: To deter fraud and counterfeiting, transactions and the creation of new currency are encrypted.
- *Limited Supply*: The number of coins that can be produced is limited, supporting the value of the cryptocurrency.

3.4.2 Uses of Cryptocurrencies

Cryptocurrencies offer a way for people to make anonymous purchases, which makes cryptocurrencies a popular vehicle for illegal transactions. Since cryptocurrencies are not connected to bank accounts, they are more difficult for law enforcement agencies to seize [37].

Cryptocurrencies are popular in Africa, Southeast Asia, and Latin America as a way for overseas workers to transfer funds back to their families [38]. Two countries have recognized cryptocurrencies as legal tender: El Salvador in 2021 and the Central African Republic in 2022.

The most popular use of cryptocurrencies is as an investment vehicle. Many people engage in speculation, purchasing cryptocurrencies in the hope that their value will rise. Some people see cryptocurrencies as a hedge against inflation.

3.4.3 Sustainability Concerns

Cryptocurrencies are supported by decentralized networks of computers. Because there is no central authority, the group of computers must agree by consensus about all cryptocurrency transactions. Distributed computers reach consensus using a formal protocol.

The encrypted, distributed ledger of cryptocurrency transactions is called a **blockchain**. **Proof of work** is a consensus protocol used to add blocks to a blockchain. To deter an attacker from creating a false blockchain, the proof-of-work protocol requires a large amount of cryptographic work to be performed to create a new block. **Block miners** running hash algorithms on special-purpose computers compete to generate the next block. When one block miner discovers a valid hash, other miners in the network verify the hash is correct. Discovering the hash takes a tremendous

computational effort; verifying the hash is correct is quick [39]. The winning block miner is paid for their effort in the virtual currency plus fees for the transactions in the new block [40]. Bitcoin and most other cryptocurrencies rely upon the proof-of-work protocol to maintain their blockchains.

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In the case of Bitcoin, a new block is added to the blockchain about once every 10 minutes. The amount of electricity used to generate a new block is astonishing. A single Bitcoin transaction has a carbon footprint equal to the carbon footprint of more than 750,000 Visa credit-card transactions. The annual electrical consumption associated with all Bitcoin transactions is comparable to the electrical consumption of the Netherlands [41]. As one of its efforts to save energy, China took steps in 2021 to end cryptocurrency mining within its borders [42].

The enormous carbon footprint of cryptocurrencies that rely on the proof of work protocol has heightened interest in another consensus protocol called proof of stake. **Proof-of-stake** protocols use validators instead of miners. By “staking” some of its cryptocurrency as collateral, a validator gets a chance to earn cryptocurrency by validating some new transactions. The algorithm randomly selects validators based on how much they have staked. The more a validator stakes,

the greater its chances of being selected. When a validator validates a good transaction, it earns cryptocurrency. If a validator validates a transaction that other validators determine to be fraudulent, it may lose some of its collateral as punishment [43, 44].

Compared to proof of work, proof of stake has the advantage that it doesn't require many computational operations to validate a transaction, making it a more environmentally responsible option. The reduced computational requirements should enable cryptocurrencies using the proof-of-stake protocol to handle far more transactions per second than cryptocurrencies relying on the proof-of-work protocol. The primary disadvantage of proof of stake is that the protocol has not yet demonstrated it is as secure from attacks as the proof-of-work protocol [43, 44].

In September 2022, Ethereum, the second largest cryptocurrency by total market capitalization, switched from proof-of-work mining to proof-of-stake mining, reducing its electrical consumption by an estimated 99.8 percent [45].

3.4.4 Financial Concerns

A variety of financial concerns have been raised about Bitcoin

and other cryptocurrencies.

Bitcoin is the dominant cryptocurrency, but only 3–7 Bitcoin transactions per second can be performed worldwide. In comparison, VisaNet can process more than 24,000 Visa transactions per second [46]. There is no way Bitcoin will ever replace credit cards for everyday transactions.

Cryptocurrency prices are volatile, making them a highly speculative investment vehicle. For example, the price of one Bitcoin was about \$1,000 in April 2017, \$65,000 in November 2021, \$16,000 in November 2022, and \$70,000 in March 2024.

Many Nobel-prize-winning economists have declared Bitcoin to be an economic bubble. Angus Deaton said, “The only advantage as far as I can see is you can be a crook. You don’t want to leave any record. It’s not an accident that people who demand ransoms demand them in Bitcoin” [47].

Since the launch of Bitcoin in 2009, over 2,000 other cryptocurrencies have failed. Many of these currencies were outright frauds or scams. Others were based on poor business plans [48]. If a cryptocurrency exchange goes bankrupt, no government protects the cryptocurrency deposits. In

November 2022 the cryptocurrency exchange FTX Trading filed for bankruptcy [49].

The proof-of-work protocol is not immune from cyberattacks. If a hacker gains control of a majority of a system's hash rate in a so-called **51% attack**, the hacker can alter events and insert double transactions, allowing the same digital coin to be spent more than once. The Digital Currency Initiative at MIT has detected dozens of 51% attacks on various cybercurrencies [50].

It is unclear how long governments will refrain from regulating cryptocurrencies. China was the largest cryptocurrency market, but in September 2021 the government of China made all cryptocurrency transactions illegal [51].

3.5 Text Messaging

Text messaging greatly increases the versatility of cell phones as information-sharing platforms. Some of the most impressive uses of text messaging are in countries where people do not have easy access to the Internet, banks, and other services taken for granted by those in countries with more infrastructure.

3.5.1 Transforming Lives in Areas with Limited Infrastructure

Text- message-based services such as M-PESA in Kenya allow people in areas with limited infrastructure to save money and pay bills using their cell phones, bypassing traditional banks. To pay a bill or transfer funds to a friend, the user simply types a text message that the recipient is able to turn into cash at any M-PESA office.

For the past decade, Kenya's Agricultural Commodities Exchange has partnered with Safaricom to provide information about crop prices to farmers via a text-messaging

service. Another service, iCow, uses voice and text messaging to help dairy farmers keep track of the gestation of their cows [52].

Counterfeit medicine is a serious problem in many African and Asian countries. Ghanaian Bright Simons came up with the idea of putting scratch cards with unique codes on packages of medicine. After scratching the package to reveal the code, a customer can text the code to a designated number to learn if the drug is genuine [52].

3.5.2 Twitter

Twitter is a Web-based social networking service that allows its users to send out text messages known as **tweets**. The service is popular because people who want their friends to know what they are doing find it more convenient to post a single tweet than to type a bunch of text messages. Many people also use Twitter as a blogging tool; they make their tweets public so that anyone can read them. Other Twitter members never post tweets, but they sign up to follow the tweets posted by other people they are interested in.

More than 200 million people use Twitter regularly [53]. Users posted a record 580,166 tweets per minute during the

World Cup Final soccer match between Germany and Brazil in 2014 [54].

3.5.3 Business Promotion

When carpenter Curtis Kimball started a part-time business running a crème brûlée cart in San Francisco, he used Twitter to let people know the cart's location and the flavors of the day. Before long, he had attracted 5,400 followers. Business became so good he quit his day job in order to keep up with demand. Many tiny businesses with no money for advertising rely upon Twitter as their only marketing tool [55].

3.6 Political Impact of Social Media and Online Advertising

3.6.1 Political Activism

Text messaging played an important role in the ousting of the president of the Philippines, Joseph Estrada, in 2001. During his impeachment trial, his political allies in the Philippine congress voted to keep some evidence against him from being revealed. Filipinos who hoped to see President Estrada convicted used text messaging to organize a demonstration on Epifanio de los Santos Avenue in Manila. Over the next couple of days, millions more text messages were sent and forwarded—many reading “Go 2 EDSA. Wear blk.”—and the crowd rose to more than one million people. Intimidated by the size of the protest, Estrada’s supporters in the Philippine congress changed their votes and allowed the incriminating evidence against President Estrada to be released. Within hours, Estrada’s presidency was over [56].

Ten years later, Twitter and Facebook played a highly visible role in the “Arab Spring” demonstrations that led to revolutions in Tunisia and Egypt, a civil war in Libya, and protests in many other Arab countries. In the midst of the protests in Cairo in 2011 that led to the resignation of President Hosni Mubarak, one protester tweeted, “We use Facebook to schedule the protests, Twitter to coordinate, and YouTube to tell the world” [57]. Arab news organization Al Jazeera created a “Twitter Dashboard” indicating the level of tweeting activity in many Arab nations where there was unrest [58].

Scholars of the Arab Spring uprisings point to an interesting phenomenon: People started using online social networks such as Twitter in order to keep up with their friends, but these interactions caused them to become politicized. Through these networks, bloggers met new people, became exposed to new ideas, and developed an interest in human rights [59].

Others think the role of social media in catalyzing social change has been overblown. They argue that social networks like Twitter and Facebook are great at building networks of people with weak connections to each other, but high-risk

activism requires strong ties among the members of a hierarchical organization [60].

The Arab Spring uprisings demonstrate how Twitter can support grassroots activism. However, Twitter can also be used by a single person or small organization to give the false impression of grassroots support of a political candidate and to trick the Google search engine into prominently featuring misinformation. Here's an early example of such an effort. In 2010 Massachusetts held a special election to fill its vacant US Senate seat. Someone supporting the Republican candidate created nine fake Twitter accounts and used them to send 979 tweets to 573 users in about two hours. Each of these tweets included the URL of a Web site containing misinformation about the Democratic candidate. These tweets were retweeted 143 times and reached 61,732 Twitter users. Google's real-time search engine detected the rapidly growing interest in the URL and promoted the URL to the top of its results page for the Democratic candidate. In other words, for a period of time anyone who typed the name of the Democratic candidate into the Google search engine received as the top result the URL of the Web page containing misinformation about that candidate [61].

Social media played a significant role in the protests following the murder of George Floyd in Minneapolis, Minnesota, on May 25, 2020. Police officers had arrested Mr. Floyd for allegedly passing a counterfeit \$20 bill in a convenience store [62]. When 17-year-old bystander Darnella Frazier saw police officers pinning Mr. Floyd to the ground outside the store, she began filming the incident on her smartphone [63]. The video clearly showed white police officer Derek Chauvin choking Mr. Floyd, who was Black, by putting his knee on Mr. Floyd's neck until Mr. Floyd was dead. The heartbreaking video, posted to Facebook, prompted Black Lives Matter protests across the United States. Between 15 and 26 million Americans marched in these protests, making it the largest protest movement in US history [64]. All four Minneapolis police officers were fired. Mr. Chauvin was convicted of murder and sentenced to 22.5 years in prison. The other three officers received prison sentences of several years [62].

Social media played an important role in the events surrounding the January 6, 2021, occupation of the US Capitol by protesters objecting to the outcome of the 2020 US presidential election. The Republican candidate was President Donald Trump, and the Democratic candidate was Joe Biden. Early in the morning of November 4, while votes

were still being counted, President Trump tweeted that Democrats were “trying to STEAL the Election” [65]. An hour-and-a-half later, he called the election results fraudulent [65]. Later that day, the first Stop the Steal group was formed on Facebook. The group quickly grew to 320,000 followers before Facebook deleted it [66]. Facebook ended up shutting down hundreds of similar groups. Eventually, pro-Trump supporters began shifting their activities to more permissive social media sites, such as Parler and Gab. Those two sites became the center of the movement to organize a protest on January 6, 2021, the date Congress was scheduled to certify the election results [66]. President Trump made several tweets between late December and early January, encouraging people to attend a protest in Washington, DC, on January 6 [66]. Right-wing groups used social media to communicate plans regarding bringing weapons into Washington, DC, and penetrating the Capitol Building [67]. At noon on January 6, President Trump held a “Save America” rally during which he called on his supporters to go to the Capitol [68]. Thousands of protesters marched to the Capitol. Between 2,000 and 2,500 people crossed the police line, entered the Capitol, and disrupted the vote certification process for hours [69]. Eventually, about 700 people were charged with federal crimes for their participation in the riot. Ironically, FBI agents used social media posts to identify many of the rioters,

and some rioters received harsher sentences because of self-incriminating evidence they had posted on social media [70].

3.6.2 Macedonian Entrepreneurs

Among US corporations, Meta (which owns Facebook) and Google are the two giants in online advertising revenue [71]. Facebook is popular among advertisers because it has a huge user base and allows advertisers to target their ads to categories of Facebook users based on various demographics, such as age, income level, locale, religion, sexual orientation, and political leanings.

Google's digital advertising offerings include contextual advertising, which places advertisements on Web pages if the content of those pages is algorithmically determined to be closely related to the advertisement. Through Google's AdSense program, people can make money from their Web pages by allowing Google to place advertisements in them. Google determines the advertisements that are displayed, and the Web site owner receives a monthly check from Google based on the number of people who clicked on the ads.

Both Meta and Google make money when people click on links leading to pages containing advertising. “Nothing drives clicks better than when the headline is exactly what people want to hear or believe,” says Ian Schafer, founder of the digital advertising agency Deep Focus [72]. Capitalists hoping to share in the advertising revenue stream know this too.

As mentioned at the beginning of the chapter, a group of young Macedonians leveraged Facebook and Google to make a lot of money for themselves in the run-up to the 2016 US presidential election. At one point, more than 100 pro-Trump Web sites were being managed from Veles, Macedonia. The Macedonians registered their sites with Google’s AdSense program, then made them attractive to visitors by copying stories from conservative Web sites in the United States, adding their own sensationalized headlines, and using Facebook’s “share” feature to drive traffic to their pages. When people clicked on their stories, Google paid them for the ads placed on their pages. The most popular stories, such as the false account that Pope Francis had endorsed Donald Trump for president, were shared by hundreds of thousands of Facebook users. Web site curators who got in on the action early made up to \$5,000 per month, a substantial sum in Macedonia [4].

3.6.3 Internet Research Agency

The Internet Research Agency (IRA), based in St. Petersburg, Russia, employs hundreds of people who use social media to exert political influence in other countries. In February 2018, a grand jury for the District of Columbia returned a 37-page indictment charging the IRA, two other Russian organizations, and 13 Russian nationals (ten of them employed by the IRA) with interfering in the US presidential election of 2016, in violation of US law. According to the indictment, the Russians set up social media accounts in the United States using false US personas. Posing as Americans, the Russians posted inflammatory material to social media sites and organized rallies with the intent of helping the campaign of Donald Trump and harming the campaign of Hillary Clinton [73].

The grand jury indictment provided evidence that in addition to the short-term goal of promoting Donald Trump and undermining Hillary Clinton, the IRA's long-term goal was to increase political polarization in the United States.

According to the indictment, after Donald Trump was elected president, IRA employees helped organize both pro-Trump

and anti-Trump rallies for the same day in New York City [73].

About a year after the election, Facebook released an estimate that the IRA published about 80,000 posts that reached the news feeds of about 29 million American Facebook users between 2015 and 2017. Another 97 million Americans received the posts after other Facebook users clicked “like” or “share” or commented on Russian-created posts. About 10 million Americans saw advertisements purchased by the IRA [74].

Since the election, many have asked whether the “fake news” stories posted by the IRA were responsible for Donald Trump’s victory. One study concluded that “fake news most likely did have a substantial impact on the voting decisions of a strategically important set of voters” [75], but another study concluded that social media may have played only a marginal role in influencing voters [76].

3.6.4 Is Democracy Being Threatened?

Whether or not the Russian misinformation campaign changed the outcome of the 2016 US presidential election, is

it possible that the widespread use of social media is threatening the pillars of democratic societies by undermining trust, “informed dialogue,” and “a shared sense of reality” [77]? In its quest to maximize its revenue, Meta wants to keep its Facebook users engaged. That means feeding them content they are interested in reading or viewing. To do this, Facebook builds profiles of user preferences and then feeds news stories and other data to users aligning with their preferences. The result is the segregation of Facebook users into ideological “echo chambers” [77].

Two-thirds of US adults read news stories on social media sites, and Facebook is by far the most popular of them, providing nearly half of US adults with at least some of their news [78]. Facebook’s algorithms help ensure people will get news stories that confirm their views, rather than expose them to alternative narratives.

Nearly two decades ago, Cass Sunstein argued that information technology could weaken democracy by allowing people to filter out news that contradicts their view of the world [79]. Today, Sunstein’s words seem prophetic. However, in one crucial respect the environment is even more insidious than what Sunstein described. Sunstein assumed that people would be actively choosing to view Web sites or

watch cable channels that confirmed their opinions. In the world moderated by Facebook, people aren't actively making the choices; instead, they are passively receiving the content delivered to them by Facebook's algorithms.

3.6.5 Troubling Times for Traditional Newspapers

The rapid growth in advertising through social media and the popularity of Craigslist have weakened the financial health of traditional print newspapers. “Craigslist came up with a new model for classified ads—free—with which newspapers could not compete” [80]. Advertising revenues for print newspapers dropped from \$44.9 billion in 2003 to \$16.4 billion in 2014 [81]. Although newspapers began selling digital advertisements, their revenues of \$3.5 billion in 2014 from online ads did not come close to making up for the \$28.5 billion decline in revenues from printed advertisements they had experienced [81].

Newspapers that had been making lots of money in prior decades began hemorrhaging cash in the twenty-first century. Many went out of business, and many others lost most of their value. In the 1990s the *New York Times* bought the *Boston Globe* for \$1.1 billion and the *Worcester Telegram &*

Gazette for \$295 million. In 2013 the *New York Times* sold both newspapers to Boston Red Sox owner John W. Henry for \$70 million, or 95 percent less than it had paid for them [82]. Two Philadelphia newspapers lost about 90 percent of their sales value between 2006 and 2012 [82].

The weakening of newspapers is likely to continue.

Newspapers continue to get about 20 percent of advertising revenue, even though they account for only about 5 percent of the time Americans spend on media. There is a good chance advertisers will continue to shift their spending away from print media to ads placed on mobile devices [83].

Are newspapers necessary to the healthy functioning of a democracy? The Founding Fathers of the United States were firm believers in the role newspapers could play in ensuring an informed citizenry and holding the powerful to account. Thomas Jefferson wrote, “[W]ere it left to me to decide whether we should have a government without newspapers or newspapers without a government, I should not hesitate a moment to prefer the latter” [84]. The power of the press, often called the Fourth Estate, was demonstrated repeatedly in twentieth-century America. Here are three illustrative examples. In the early 1900s investigative journalist Ida Tarbell’s exposé of John D. Rockefeller and the Standard Oil

Company helped lead to the determination that Standard Oil was violating antitrust laws and the subsequent breakup of the company. In 1971 the *New York Times* published the Pentagon Papers, revealing a wealth of information about the conduct of the Vietnam War during the Kennedy and Johnson administrations that had been kept secret from the American public. Bob Woodward and Carl Bernstein's investigation of the White House's involvement in the break-in of the Democratic National Headquarters at the Watergate Hotel in 1972 led to the resignation of President Richard Nixon in 1974.

To be fair, there is plenty of evidence that Americans' interest in newspapers was ebbing well before the creation of Craigslist and the start of the online advertising revolution. Newspaper circulation per capita has been steadily declining since the late 1940s [85]. However, until the turn of the century the best newspapers continued to make healthy profits by introducing automation to make producing and printing newspapers less labor-intensive, consolidating with other newspapers (reducing competition), and steadily increasing their advertising rates. What has proven to be a much more difficult challenge for newspapers to overcome has been devising a strategy to remain profitable in an environment of steadily declining advertising revenues.

Are print newspapers facing extinction in the United States?
If so, can the Fourth Estate retain its vibrancy, power, and
independence without traditional newspapers?

3.7 Censorship

Censorship is the attempt to suppress or regulate public access to material considered offensive or harmful.

Historically, most censorship has been exercised by governments and religious institutions. For example, Roman censors banished the poets Ovid and Juvenal for their writings. During the Middle Ages the Inquisition suppressed the publication of many books, including the work of Galileo Galilei.

Censorship became a much more complicated issue with the invention of the printing press. The printing press broke the virtual monopoly held by governments and religious institutions on distributing material to a large audience, and the increase in printed material resulted in a greater number of literate people. For the first time, private individuals could broadcast their ideas to others on a wide scale.

In many countries, religious institutions continue to play a significant role in determining what material should be accessible to the public. In Western democracies, the gradual separation of church and state has left the government as the sole institution responsible for censorship.

3.7.1 Direct Censorship

Direct censorship has three forms: government monopolization, prepublication review, and licensing and registration.

The first form of direct censorship is government monopolization. In the former Soviet Union, for example, the government owned all the television stations, radio stations, and newspapers. Private organizations could not even own a photocopy machine. Government monopolization is an effective way to suppress the flow of information. Modern computer and communication technology makes government monopolization much more difficult than in the past.

Prepublication review is the second form of direct censorship. This form of censorship is essential for material the government wishes to keep secret, such as information about its nuclear weapons program. Most governments have laws restricting the publication of information that would harm national security. In addition, autocratic governments typically block publication of material deemed injurious to the reputations of their rulers.

The third form of direct censorship is licensing and registration. This form of censorship is typically used to control media with limited bandwidth. For example, only a limited number of radio and television stations can be accommodated on the electromagnetic spectrum. Hence a radio or television station must obtain a license to broadcast at a particular frequency. Licensing invites censorship. For example, the US Federal Communications Commission has banned the use of certain four-letter words. This led to a challenge that went all the way to the US Supreme Court, as we see in [Section 3.8.3](#).

3.7.2 Self-Censorship

Perhaps the most common form of censorship is self-censorship: a group deciding for itself not to publish material. In some countries a publisher may censor itself in order to avoid persecution. For example, after US-led forces toppled the regime of Saddam Hussein in April 2003, CNN's chief news executive, Eason Jordan, admitted that CNN had suppressed negative information about the actions of the Iraqi government for more than a decade in order to keep CNN's Baghdad bureau open and protect Iraqi employees of CNN [86].

In other countries, publishers may want to maintain good relations with government officials. Publications compete with each other for access to information. Often this information is available only from government sources. Publishers know that if they offend the government, their reporters may not be given access to as much information as reporters for rival publications, putting them at a competitive disadvantage. This knowledge can lead a “free” press to censor itself.

Publishers have adopted ratings systems as a way of helping people decide if they (or their children) should access particular offerings. For example, television stations in the United States broadcast shows with “mature content” late in the evening. Voluntary ratings systems help people decide if they (or their children) will see a movie, watch a television show, or listen to a CD.

The Web does not have a universally accepted ratings system. Some Web sites practice a form of labeling. For example, the home page may warn the user that the site contains nudity and require the user to click on an “I agree” button to enter the site. However, other sites have no such warnings. People who stumble onto these sites are immediately confronted with images and text they may find offensive.

3.7.3 Challenges Posed by the Internet

Five characteristics of the Internet make censorship more difficult:

1. *Unlike traditional one-to-many broadcast media, the Internet supports many-to-many communications.*

While it is relatively easy for a government to shut down a newspaper or a radio station, it is more difficult for a government to prevent an idea from being published on the Internet, where millions of people have the ability to post Web pages.

2. *The Internet is dynamic.*

Millions of new devices are being connected to the Internet each year.

3. *The Internet is huge.*

There is simply no way a team of human censors can keep track of *everything* that is posted on the Web.

While automated tools are available, they are fallible.

Hence any attempt to control access to material stored on the Internet cannot be 100 percent effective.

4. *The Internet is global.*

National governments have limited authority to restrict activities happening outside their borders.

5. *It is hard to distinguish between children and adults on the Internet.*

How can an “adult” Web site verify the age of someone attempting to enter the site?

3.7.4 Government Filtering and Surveillance of Internet Content

Despite the difficulties facing those who would seek to censor Internet content, studies reveal that governments around the globe are in fact limiting access to the Internet in a variety of ways [87].

One approach is to make the Internet virtually inaccessible. In North Korea, for example, only about 20,000 people have access to the Internet, representing 0.1% of the population [88].

In other countries, Internet access is easier but still carefully controlled. For example, the Internet backbone in Saudi Arabia is owned by the government, and all Internet traffic to and from the rest of the world flows through two gateways, which block pornography sites, gambling sites, and many other pages deemed to be offensive to Islam or the government of Saudi Arabia [89].

The Chinese government has blocked access to the Internet during times of social unrest. For example, in July 2009, China responded to ethnic riots in the autonomous region of Xinjiang by turning off Internet service to the entire region for 10 months [90, 91].

In addition, China has built one of the world's most sophisticated Web filtering systems [92]. The Great Firewall of China prevents Chinese citizens from accessing certain Internet content by blocking messages coming from blocklisted sites, which include Google, YouTube, and Facebook [93]. According to Freedom House, the Chinese Communist Party's Central Propaganda Department employs "hundreds of thousands or even millions of people to monitor, censor, and manipulate online content" [93]. The government shuts down Web sites and censors blogs that criticize government policy, discuss politically sensitive

topics, or provide information that puts the government in a bad light. Among the Web sites blocklisted by the government include those containing pornography, those associated with the Dalai Lama or the Falun Gong, those referring to the 1989 military crackdown, and those run by certain news organizations, such as Voice of America and BBC News. Before the 2008 Summer Olympics, the International Olympic Committee assured journalists that they would have unfettered access to the Internet during their stay in Beijing, but once the journalists arrived in Beijing, they discovered that many sites were blocked. The International Olympic Committee admitted that it had agreed to allow the Chinese government to block sensitive sites “not considered Games related” [94]. The Chinese government is also trying to find ways to regulate access to virtual private networks, which many Chinese people use to circumvent the Great Firewall [93].

Meanwhile, Western nations have different standards about what is acceptable and what is not. For example, Germany forbids access to any neo-Nazi Web site, but Web surfers in the United States can access many such sites.

Political satire and pornography are easily available through American ISPs. Americans are used to political satire, but

many citizens are concerned about the corrupting influence of pornography, particularly on minors. Since 1996 the US Congress has passed three laws aimed at restricting children's access to sexually explicit material on the Web: the Communications Decency Act, the Child Online Protection Act, and the Children's Internet Protection Act. The first two laws were ruled unconstitutional by the US Supreme Court; the third was upheld by the Supreme Court in June 2003.

3.7.5 Ethical Perspectives on Censorship

It is interesting that Immanuel Kant and John Stuart Mill, who had quite different ethical theories, had similar views regarding censorship.

Kant's Views on Censorship

As a thinker in the tradition of the Enlightenment, Kant's motto was, "Have courage to use your own reason" [95]. Kant asks the rhetorical question, "Why don't people think for themselves?" and answers it: "Laziness and cowardice are the reasons why so great a portion of mankind, after nature has long since discharged them from external direction, nevertheless remain under lifelong tutelage, and why it is so

easy for others to set themselves up as their guardians. It is so easy not to be of age. If I have a book which understands for me, a pastor who has a conscience for me, a physician who decides my diet, and so forth, I need not trouble myself. I need not think, if I can only pay—others will readily undertake the irksome work for me” [95, p. 85].

5 The Enlightenment was a reaction to the institutional control over thought held by the aristocracy and the Church. Kant believed he was living in a time in which the obstacles preventing people from exercising their own reason were being removed. He opposed censorship as a backward step.

Mill's Views on Censorship

John Stuart Mill also championed freedom of expression. He gave four reasons why freedom of opinion, and freedom of expression of opinion, were necessary.

First, none of us is infallible. All of us are capable of error. If we prevent someone from voicing their opinion, we may actually be silencing the voice of truth.

Second, while the opinion expressed by someone may be erroneous, it may yet contain a kernel of truth. In general, the

majority opinion is not the whole truth. We ought to let all opinions be voiced so that all parts of the truth are heard.

Third, even if the majority opinion should happen to be the whole truth, it is in the clash of ideas that this truth is rationally tested and validated. The whole truth left untested is simply a prejudice.

Fourth, an opinion that has been tested in the fire of a free and open discourse is more likely to have a “vital effect on the character and conduct” [96, p. 61].

Therefore, Mill, like Kant, fundamentally supported the free exchange of ideas, with the conviction that good ideas would prevail over bad ones. Applying their philosophy to the World Wide Web, it seems they would support the free exchange of opinions and oppose any kind of government censorship of opinions.

Mill's Principle of Harm

However, a lack of government censorship can also lead to harm. Under what circumstances should the government intervene? Mill proposed the principle of harm as a way of deciding when an institution should intervene in the conduct of an individual.

Principle of Harm

The only purpose for which power can be rightfully exercised over any member of a civilized community, against his will, is to prevent harm to others. His own good, either physical or moral, is not a sufficient warrant [96, p. 12].

In other words, the government should not get involved in the private activities of individuals, even if the individuals are doing something to harm themselves. Only if individuals' activities are harming other people should the government step in.

The principle of harm can be used to explain the position of most Western democratic governments with respect to censoring pornographic material depicting adults. Some ethicists conclude it is not wrong for adults to view pornography depicting adults. Others hold that this activity is immoral. If the activity is immoral, it is more certain the harm is being done to the individual consumer; less certain is

how much harm is being done to other people. Hence the principle of harm can be used as an argument why the government should not be trying to prevent adults from using pornography depicting adults.

3.8 Freedom of Expression

In the United States, freedom of expression is one of the most cherished—and most controversial—rights. In this section we explain the history behind the adoption of the First Amendment to the United States Constitution. We also explore why the freedom of expression has not been treated as an absolute right.

3.8.1 History

At the time of the American Revolution, any criticism of government was seen as a threat to public order and could result in fines and/or imprisonment. Restrictions on freedom of speech in England date back to 1275 and a law called De Scandalis Magnatum. According to this law, a person could be imprisoned for spreading stories about the King that could have the effect of weakening the loyalty of his subjects. The scope of the law became much broader through numerous revisions over the next two centuries. Eventually, it

encompassed seditious words and words spoken against a wide variety of government officials, including justices [97].

De Scandalis Magnatum was administered by the Court of Star Chamber, or “Star Chamber” for short. The Star Chamber reported directly to the King, and it did not have to obey traditional rules of evidence. Rulings of the Star Chamber demonstrated that a person could be convicted for making a verbal insult or for something written in a private letter. The Star Chamber was abolished in 1641, but the law continued to be enforced through Common Law courts [97].

At the end of the eighteenth century, freedom of the press in England and its colonies meant freedom to print without a license. In other words, there were no **prior restraints** on publication. People could publish what they pleased.

However, those who published material found to be seditious or libelous would face severe consequences [97].

The law against libel simply considered if the material printed was harmful; arguing that the information was true was not relevant to the proceedings and could not be used in a publisher’s defense. Between 1760 and the end of the American Revolution, about 50 people were successfully prosecuted for libel. To prevent such prosecutions from

continuing, most states adopted bills of rights after gaining independence from England [97].

In May 1787, delegates from the thirteen states gathered in Philadelphia to revise the Articles of Confederation. Soon they were drafting a completely new Constitution. Delegate George Mason, author of the Virginia Declaration of Rights, strongly opposed the proposed Constitution because it contained no declaration of the rights of the citizens. Patrick Henry and other political leaders shared Mason's objections [97].

While the proposed Constitution was ratified by all thirteen states, most state legislatures adopted the Constitution with the expectation that Congress would offer amendments addressing the human-rights concerns brought up by the opponents of the Constitution. During the first Congress, James Madison proposed 12 such amendments. All 12 of these amendments were sent to the states for ratification. Of these 12 amendments, 10 were quickly ratified. Today these 10 amendments are commonly known as the Bill of Rights. The first of these amendments, the one Madison considered most essential, was the one guaranteeing freedom of speech and freedom of the press [97].

First Amendment to the United States Constitution

Congress shall make no law respecting an establishment of religion, or prohibiting the free exercise thereof; or abridging the freedom of speech, or of the press; or the right of the people peaceably to assemble, and to petition the government for a redress of grievances.

3.8.2 Freedom of Expression Not an Absolute Right

The primary purpose of the First Amendment's free speech guarantee is political. Free speech allows an open discussion of public issues. It helps make government responsive to the will of the people [98].

However, the First-Amendment right to free expression is not limited to political speech. Nonpolitical speech is also covered. There are good reasons for protecting nonpolitical as well as political speech. First, it is sometimes hard to draw the line between the two. Asking a judge to make the distinction turns it into a political decision. Second, society can benefit from nonpolitical as well as political speech. Hence the free-speech guarantee of the First Amendment also promotes scientific and artistic expression. For the same reason, the definition of “speech” encompasses more than words. Protected “speech” includes art and certain kinds of conduct, such as burning a US flag [99].

Decisions by the US Supreme Court have made clear that freedom of expression is not an absolute right. Those who abuse this freedom and harm the public may be punished. For example, protection is not given to “libel, reckless or calculated lies, slander, misrepresentation, perjury, false advertising, obscenity and profanity, solicitation of crime, and personal abuse or ‘fighting’ words,” because these actions do not serve the ends of the First Amendment [98].

Various restrictions on freedom of speech are justified because of the greater public good that results. For example, US law prohibits cigarette advertising on television because

cigarette smoking has detrimental effects on public health. Some cities use zoning laws to concentrate adult bookstores in a single part of town because the presence of adult bookstores lowers property values and increases crime.

3.8.3 FCC v. Pacifica Foundation

To illustrate limits to First Amendment protections, we consider the decision of the US Supreme Court in the case of *Federal Communications Commission v. Pacifica Foundation et al.*

In 1973 George Carlin recorded a performance made in front of a live audience in California. One track on the resulting record is a 12-minute monologue called “Filthy Words.” In the monologue Carlin lists seven words that “you couldn’t say on the public, ah, airwaves, um, the ones you definitely wouldn’t say, ever” [100]. The audience laughs as Carlin spends the rest of the monologue creating colloquialisms from the list of banned words.

On the afternoon of October 30, 1973, counterculture radio station WBAI in New York aired “Filthy Words” after warning listeners the monologue contained “sensitive

language which might be regarded as offensive to some” [101]. A few weeks after the broadcast, the Federal Communications Commission (FCC) received a complaint from a man who had heard the broadcast on his car radio in the presence of his son. In response to this complaint, the FCC issued a declaratory order and informed Pacifica Foundation (the operator of WBAI) that the order would be placed in the station’s license file. The FCC warned Pacifica Foundation that further complaints could lead to sanctions.

Pacifica sued the FCC, and the resulting legal battle reached the US Supreme Court. In 1978 the Supreme Court ruled, in a 5–4 decision, that the FCC did not violate the First Amendment [101]. The majority opinion states, “Of all forms of communication, it is broadcasting that has received the most limited First Amendment protection.” There are two reasons why broadcasters have less protection than booksellers or theater owners:

1. *“Broadcast media have a uniquely pervasive presence in the lives of all Americans.”* [101]

Offensive, indecent material is broadcast into the privacy of citizens’ homes. Since people can change stations or turn their radios on or off at any time, prior warnings cannot completely protect people from being

exposed to offensive material. While someone may turn off the radio after hearing something indecent, that does not undo a harm that has already occurred.

2. *“Broadcasting is uniquely accessible to children, even those too young to read.”* [101]

In contrast, restricting children’s access to offensive or indecent material is possible in bookstores and movie theaters.

The majority emphasized that its ruling was a narrow one and that the context of the broadcast was all-important. The time of day at which the broadcast occurred (2 p.m.) was an important consideration, because that affected the composition of the listening audience.

3.8.4 Case Study: Leah’s Blog

Leah is a journalism major who maintains a popular blog focusing on campus life. Leah attends a private birthday party in someone’s apartment for her friend Jamal, a college student active in the Whig Party on campus. Someone gives Jamal a Tory Party T-shirt as a gag gift, and Jamal puts it on. Leah uses her cell phone to get a picture of Jamal wearing the

T-shirt when he is looking the other way. Jamal gives Leah a ride home after the party, but she does not tell him about the photo. When she is back in her apartment, she posts the photo on her blog. In the blog she identifies Jamal and explains the context in which the photo was taken.

The story is read by many people both on and off campus. The next day Jamal confronts Leah, yells at her for posting the photo, and demands that she remove it from her Web site. Leah complies with Jamal's request by removing the photo, and the two of them remain friends. As a result of the incident, Jamal becomes more popular on campus, and the number of people who read Leah's blog increases.

) Was it wrong for Leah to post the picture of Jamal on her blog without first getting his permission?

Kantian Analysis

By uploading Jamal's photo to her blog without first asking his permission, Leah didn't respect Jamal's autonomy.

Instead, she treated him as a means to her end of increasing the readership of her blog. Therefore, her action was wrong according to the second formulation of the Categorical Imperative.

Social-Contract-Theory Analysis

The birthday party was held in the apartment of one of Jamal's friends. In this private setting and among friends, Jamal had a legitimate expectation that what happened during the party would not be broadcast to the world. By secretly taking a photo of Jamal doing something out of character and posting that photo on her blog, Leah violated Jamal's right to privacy. For this reason Leah's action was wrong.

Act-Utilitarian Analysis

We need to determine the positive and negative consequences of Leah's action on the two people involved. Leah increased the popularity of her blog, which is precisely the positive outcome she wanted (+10). Jamal's anger at Leah shows that he was hurt and upset by what she did, but after he confronted her, she removed the photo from her Web site and they reconciled. Therefore, while the intensity of this negative consequence to Jamal was significant, its duration was brief (−5). As a result of the posting, Jamal became more popular on campus, a very good thing for someone active in campus politics (+10). Jamal had Leah to thank for this boost in his popularity, further quenching the unhappiness he initially felt when he learned what she had done (+2). We

conclude that the short-term consequences for both Leah and Jamal were positive (totaling +17).

The long-term consequences are difficult to determine. It is possible that the photo could land in the wrong hands and be used to discredit Jamal someday in the future (−100), but this would depend on many factors. Jamal is currently politically active. Is he going to stay active in Whig politics after he graduates from college (50 percent)? The photo was only on the Web for a day. Did anyone download it (20 percent)? If so, what is the chance that someday the photo will fall into the hands of someone who wants to make Jamal look bad (10 percent)?

An important part of a utilitarian analysis is looking at the certainty of each consequence: in other words, the probability that it will happen. The short-term consequences of Leah's action are certainly positive for both Leah and Jamal ($+17 \times 100\% = +17$). The long-term negative consequences, if any, are not certain at all ($-100 \times 50\% \times 20\% \times 10\% = -1$). Adding the calculated short-term benefits (+17) to the calculated long-term harms (−1) yields a total benefit of 16, and we conclude her action had a good outcome.

Rule-Utilitarian Analysis

Let's consider what would happen if everyone were constantly taking photos of everyone they bumped into and posting them on the Web. There would be some positive consequences. It would be easier for people to see what their friends were up to. People might be more reluctant to engage in illegal activities if they thought photo or video evidence might appear on the Web. There would also be a variety of negative consequences. Once people started to feel as if they were always being photographed, they would become self-conscious, making it more difficult for them to simply be themselves. People would be less free to take off their public persona and express their true feelings. Inevitably, people would post photos that caused hard feelings and led to strained relationships. Ultimately, the negative consequences seem to be more weighty than the positive consequences, and we conclude Leah's action was wrong.

Virtue-Ethics Analysis

Leah and Jamal are friends. Aristotle recognized that people are social beings and that friendship plays an important role in *eudaimonia*, or human flourishing. True friends trust each other and seek each other's good. Reciprocity and an equality of interest are fundamental elements of friendship. There was no reciprocity when Leah sneakily took Jamal's photo

without his knowledge; she exploited him by taking something from him without giving him anything in return. She did not act as a friend when she put her own interest above that of Jamal. After the party, Leah had another opportunity to be honest with Jamal. Unfortunately, she did not trust Jamal enough to admit she had taken the picture and ask his permission before posting the photo. Instead, she said nothing to Jamal and simply posted the photo to her blog. Overall, Leah's actions at several points in the story do not seem to be characteristic of a good friend.

Summary

The analyses from the perspectives of Kantianism, social contract theory, rule utilitarianism, and virtue ethics do not support Leah posting the photo without asking Jamal's permission, though each analysis uses a different line of reasoning to reach that conclusion. Leah imagined (correctly, as it turns out) that Jamal would be angry if she took a photo of him wearing the Tory Party T-shirt, and that is why she took the photo when he wasn't looking. Leah figured it would be better to beg for forgiveness than ask for permission, but what she did was cut Jamal out of a decision that affected both of them. This is no way to treat anybody, much less a friend. Leah would have been better off trying to

persuade Jamal that putting the photo on her blog would be to their mutual advantage, posting the image only after obtaining his consent.

3.9 Children and Inappropriate Content

Many parents and guardians believe they ought to protect their children from exposure to pornographic and violent materials. A few years ago the center of concern was the Web, and a large software industry sprang up to provide browsers with the ability to block inappropriate images. Now smartphones are ubiquitous, and some caregivers are being forced to confront the unpleasant reality that children have texted unclothed or sexually explicit images of themselves to friends or even strangers.

3.9.1 Web Filters

A **Web filter** is a piece of software that prevents certain Web pages from being displayed by your browser. While you are running your browser, the filter runs as a background process, checking every page your browser attempts to load.

If the filter determines that the page is objectionable, it prevents the browser from displaying it.

Filters can be installed on individual computers, or an ISP may provide filtering services for its customers. Programs designed to be installed on individual computers, such as CyberSentinel and Spector Pro, can be set up to email caregivers as soon as they detect an inappropriate Web page [102].

Typical filters use two different methods to determine if a page should be blocked. The first method is to check the URL of the page against a blocklist of objectionable sites. If the Web page comes from a blocklisted site, it is not displayed. The second method is to look for combinations of letters or words that may indicate a site has objectionable content.

Neither of these methods is foolproof. The Web contains millions of pages containing pornography, and new sites continue to be created at a high rate, so any blocklist of pornographic sites will be incomplete by definition. Some filters sponsored by conservative groups have blocklisted sites associated with liberal political causes, such as those sponsored by the National Organization for Women and LGBTQIA+ groups. The algorithms used to identify

objectionable words and phrases can cause Web filters to block out legitimate Web pages.

3.9.2 Child Internet Protection Act

In March 2003, the Supreme Court weighed testimony in the case of *United States v. American Library Association*. The question: Can the government require libraries to install antipornography filters in return for receiving federal funds for Internet access?

More than 14 million people access the Internet through public library computers. About one-sixth of the libraries in the United States have already installed filtering software on at least some of their computers. The Child Internet Protection Act requires that libraries receiving federal funds to provide Internet access to its patrons must prevent children from getting access to visual depictions of obscenity and child pornography. The law allows adults who desire access to a blocked page to ask a librarian to remove the filter.

In his testimony before the Supreme Court, Solicitor General Theodore Olson argued that since libraries don't offer

patrons X-rated magazines or movies, they should not be obliged to give them access to pornography over the Internet.

Paul Smith, representing the American Library Association and the American Civil Liberties Union, argued that in their attempt to screen out pornography, filters block tens of thousands of inoffensive pages. He added that requiring adults to leave the workstation, find a librarian, and ask for the filter to be turned off would be disruptive to their research and would stigmatize them.

In June 2003, the US Supreme Court upheld CIPA, ruling 6–3 that antipornography filters do not violate First-Amendment guarantees [103]. Chief Justice William Rehnquist wrote, “A public library does not acquire Internet terminals in order to create a public forum for Web publishers to express themselves, any more than it collects books in order to provide a public forum for the authors of books to speak . . . Most libraries already exclude pornography from their print collections because they deem it inappropriate for inclusion” [104].

3.9.3 Ethical Evaluations of CIPA

In this section we evaluate CIPA from the perspectives of Kantianism, act utilitarianism, and social contract theory.

Kantian Evaluation

We have already covered Kant's philosophical position against censorship. He believed that allowing people to use their own reason would lead to society's gradual enlightenment. In this case, however, the focus is narrower. Rather than talking about censorship in general, let's look at CIPA in particular.

The goal of CIPA is to protect children from the harm caused by exposure to pornography. The way the goal is being implemented is through Web filters. Studies have demonstrated that Web filters do not block all pornographic material but do block some nonpornographic Web pages. Some nonpornographic information posted on the Web is not easily accessible at libraries implementing government-mandated Web filters. The people posting this information did not consent to their ideas being blocked. Hence the decision to require the use of Web filters treats the creators of nonoffensive but blocked Web pages solely as means to the end of restricting children's access to pornographic materials. This analysis leads us to conclude that CIPA is wrong.

Act-Utilitarian Evaluation

Our second evaluation of CIPA is from an act-utilitarian point of view. What are the consequences of passing CIPA?

1. While not all children access the Web in public libraries, and while Web filtering software is imperfect, it is probable that enacting CIPA results in fewer children being exposed to pornography, which is good.
2. Because Web filters are imperfect, people are unable to access some legitimate Web sites. As a result, Web browsers in libraries are less useful as research tools, a harmful consequence.
3. Adult patrons who ask for filters to be removed may be stigmatized (rightly or not) as people who want to view pornography, a harm to them.
4. Some blocked sites may be associated with minority political views, reducing freedom of thought and expression, which is harmful.

Whenever we perform the utilitarian calculus and find some benefits and some harms, we must decide how to weigh them. This is a good time to think about utilitarian philosopher Jeremy Bentham's seven attributes. In particular,

how many people are in each affected group? What is the probability the good or bad event will actually happen? How soon is the event likely to occur? How intense will the experience be? To what extent is the pain not diluted by pleasure or vice versa? How long will it last? How likely is the experience to lead to a similar experience? Actually performing the calculus for CIPA is up to each person's judgment. Different people could reach opposite conclusions about whether enacting CIPA is the right thing for the US government to do.

Social-Contract-Theory Evaluation

In social contract theory, morally binding rules are those rules mutually agreed to in order to allow social living [105]. Freedom of thought and expression is prized. According to John Rawls, "liberty of conscience is to be limited only when there is a reasonable expectation that not doing so will damage the public order which the government should maintain" [106].

It would be difficult to gain consensus around the idea that the private viewing of pornography makes social living no longer possible. For this reason, the private use of pornography is considered to be outside the social contract and nobody else's business. However, when we think about

the availability of pornography in public libraries, the issue gets thornier.

Some argue that allowing people to view pornography in a public place demeans women, denying them dignity as equal persons [107]. On the other hand, we know that filtering software is imperfect. In the past, it has been used to promote a conservative political agenda by blocking sites associated with other viewpoints [108, 109]. Hence it reduces the free exchange of ideas, limiting the freedoms of thought and expression. For some adults, public libraries represent their only opportunity to access the Web at no cost. In order to be treated as free and equal citizens, they should have the same Web access as people who have Internet access from their homes. If Web filters are in place, their access is not equal because they must ask for permission to have the filters disabled. Finally, while most people would agree that children should not be exposed to pornographic material, it would be harder to convince reasonable people that social living would no longer be possible if children happened to see pornography in a library.

Our analysis from the point of view of social contract theory has produced arguments both supporting and opposing the Children's Internet Protection Act. However, installing filters

does not seem to be necessary to preserve the public order. For this reason, the issue is outside the social contract and freedom of conscience should be given precedence.

3.9.4 Sexting

Sexting refers to sending sexually suggestive text messages or emails containing nude or nearly nude photographs [110].

Sexting has drawn a lot of media attention, but early reports on this phenomenon may have overstated the rates at which teenagers are sexting. More recent surveys indicate that between 3 and 7 percent of teens are sexting, and only 1 percent of teens have actually sent photos showing intimate areas of their bodies [111].

Whatever its frequency, sexting has had a serious impact on some people's lives. Here are three incidents.

Ohio high-school student Jesse Logan sent nude pictures of herself to her boyfriend. When they broke up, the ex-boyfriend distributed the photos to other girls in her high school. Jesse endured months of harassment from her high-school classmates and began skipping classes on a daily basis. After attending the funeral of another classmate who died by suicide, Jesse went home and ended her own life [112].

Ting- Yi Oei, a 59-year-old assistant principal at Freedom High School in South Riding, Virginia, was asked to investigate rumors that students were distributing nude photographs on their cell phones. His investigation led to a 16-year-old boy, who had a photo of a 17-year-old girl on his phone. In the photo, the girl was posing in her panties, with her arms mostly covering her breasts. Oei showed the image to the principal, who told him to keep a copy on his computer as evidence. Two weeks later the same boy got in trouble again, and Oei suspended him for two weeks. When Oei met with the boy's mother, he told her about the earlier photo incident. The boy's mother was upset that Oei hadn't immediately told her about the photo, and she demanded that Oei revoke her son's suspension. When Oei refused, the mother went to the police and told them about the photo. Sheriff's investigators came to the school and found the photo of the girl on Oei's computer. County prosecutor James Plowman gave Oei an ultimatum: resign or face felony charges for possession of child pornography. Plowman's assistant told the press, "We just feel very strongly that this is not someone who should be in the Loudoun County school system." Oei refused to resign, and in August 2008, a grand jury indicted him for possession of child pornography. The school district removed him from his position as vice

principal and reassigned him to a job at a testing center. Oei had to take out a second mortgage on his house to pay legal expenses. In April 2009, Loudoun Circuit Court Judge Thomas Horne dismissed the charges, noting that nudity alone is not sufficient to categorize an image of a minor as child pornography. Though never convicted, Oei ended up deeply in debt and with a tarnished reputation, unsure if he would ever return to his former position at the high school [113].

After Phillip Alpert got into an argument with his 16-year-old girlfriend, he emailed a nude photo of her to dozens of her friends and family members. The Orlando, Florida, police arrested Alpert, who had just turned 18, charging him with sending child pornography, a felony. Alpert was sentenced to five years' probation and required to register with the state of Florida as a sex offender. He will remain a registered sex offender until he is 43 years old [114].

There appears to be a widespread sentiment that child pornography laws should not be used to prosecute teenagers who are caught sexting. As of August 2022, 13 states in the United States had passed laws allowing minors who sext to be diverted outside the formal justice system [115].

3.10 Breaking Trust

Some people use the Internet for purposes of identity theft or sexual assault. Companies and individuals have used fake online reviews to drum up business. Many people have been humiliated by the widespread dissemination of photos or videos they expected to be kept private.

5

3.10.1 Identity Theft

Dorothy Denning defines **identity theft** as “the misuse of another person’s identity, such as name, Social Security number, driver’s license, credit card numbers, and bank account numbers. The objective is to take actions permitted to the owner of the identity, such as withdraw funds, transfer money, charge purchases, get access to information, or issue documents and letters under the victim’s identity” [116].

By far the leading form of identity theft in the United States involves the fraudulent use of an existing credit card account or bank account. In 2017 about 7 percent of adults in the United States reported being the victims of identity fraud [117]. Fortunately, US law says that a consumer’s liability for losses due to credit card fraud are limited to \$50 if reported

promptly. Most victims end up paying nothing out of pocket because their banks and credit card companies offer zero-liability fraud protection.

Point- of-sale fraud has declined rapidly with the adoption of EMV (embedded chip) credit cards and terminals. Identify fraud is increasingly happening online, where it is more difficult for merchants to verify that transactions are being authorized by the rightful cardholders because the cardholders cannot present their cards for inspection. So-called “card-not-present fraud” is now 81 percent more likely than point-of-sale fraud [117].

Identity thieves are much more likely to use low-tech methods rather than sophisticated online attacks to gain access to the account information they need. A 2008 survey of identity theft victims revealed that in 43 percent of the cases, the theft was the result of a lost or stolen wallet, credit card, checkbook, or another physical document [118]. Some identity thieves engage in **dumpster diving**— looking for personal information in garbage cans or recycling bins. Old bills, bank statements, and credit card statements contain a wealth of personal information, including names, addresses, and account numbers. Another simple way to get information

is through **shoulder surfing**—looking over the shoulders of people filling out forms.

Gathering financial information via spam is called **phishing** (pronounced “fishing”). Thieves send out spam messages designed to look as if they originated from PayPal, eBay, or another well-known Internet-active business. Through these messages they hope to con unsuspecting recipients into connecting with authentic-looking Web sites and revealing their credit card numbers or other personal information. For example, a victim might receive an email message purportedly from PayPal, asking the person to go to the PayPal Web site to confirm a transaction. The email message contains a hypertext link. When the person clicks on the link, they are connected to the counterfeit PayPal site.

College students are five times more likely to be victims of identity theft than average adults, for a variety of reasons. They use a lot of digital devices, and many do not secure them properly with strong passwords. They tend to post a lot of personal information on social media sites. College students are particularly vulnerable to low-tech methods of identity theft. A high percentage of them live in close quarters with others, and some do not take care to secure sensitive information. Dorm rooms are often left wide open, with

checkbooks and financial documents kept in unlocked desk drawers. Students may throw documents containing personal financial information into the trash, where they can be retrieved by dumpster divers [119].

The Identity Theft and Assumption Act of 1998 makes identity theft a federal crime. In 2004 Congress passed the Identity Theft Penalty Enhancement Act, which lengthened prison sentences for identity thieves [120]. A variety of law-enforcement agencies investigate alleged violations of this law: the US Secret Service, the FBI, the US Postal Inspection Service, and the Office of the Inspector General of the Social Security Administration.

7

3.10.2 Fake Reviews

A 2014 survey of North Americans by BrightLocal revealed that 88 percent had used online reviews in the past 12 months to gauge the quality of a local business, and 39 percent read reviews regularly. Restaurants, hotels, doctors and dentists, and beauty salons were the business types most frequently searched by consumers [121].

About three-quarters of consumers (72 percent) told surveyers that “positive reviews make them trust a local business more” [121]. Unfortunately, a significant percentage

of online reviews are fraudulent. Some businesses try to boost sales by posting fake positive reviews of their enterprises or fake negative reviews of their competitors. Alternatively, they may hire third parties to do these things.

One study estimated that 16 percent of restaurant reviews on Yelp were fraudulent [122]. Another researcher concluded that one-third of all Internet reviews were fake [122]. Fake reviews undermine the credibility of the information consumers are using to make purchasing decisions.

In some instances consumers can identify suspect reviews. For example, by tracing through the reviewer's alias, a consumer can see how many reviews that reviewer has written. An alias that has posted only a single review is suspicious [122].

Another way to screen reviews is to look at the number of stars. "Pretty much all fake reviews are 5-star or 1-star reviews. Nobody leaves a fake 3-star review. What would be the point?" says Nicholas White of the *Daily Dot* [123].

Online review provider Yelp invests millions of dollars annually attempting to identify fake reviews posted to its site. About 10 percent of its employees are on the hunt for fraudulent reviews, and it has also developed sophisticated software to detect suspicious reviews. About 25 percent of

the reviews submitted to Yelp don't make it past the screening and are not recommended [123].

3.10.3 Online Predators

Instant messaging refers to a system allowing two people to “chat” via typing in real time over the Internet. Instant messaging is popular among young adults because it allows them to communicate without paying SMS text-messaging charges. Kik Messenger is a popular instant messaging app for smartphones; as of January 2018, it had 300 million users, including 33 percent of the teens in the United States [124]. In addition to text messages, Kik Messenger also allows users to send images and videos.

In August 2013, police began to issue warnings about people using Kik Messenger to facilitate their perpetration of sexual abuse and assault [125]. A few months later, 39-year-old Manuel Salto pleaded guilty to third-degree criminal sexual conduct after using Kik Messenger to meet a 13-year-old girl whom he later lured to his home for sexual relations [126].

Other apps as well have been used by people who engage in pedophilia. According to police, Whisper is particularly

dangerous because it allows users to post their location, making it easier for predators to find local targets [127].

3 For more than 15 years, police have launched sting operations to arrest people seeking sex with minors. Sometimes these sting operations have resulted in dozens of arrests. For example, the Osceola County sheriff's department in Florida conducted a one-week-long sting operation in January 2012 that resulted in the arrest of 40 adults, including a professional golfer, a teacher, and a swim coach. Deputies arrested the suspects as they arrived at an undercover house, thinking they were going to have sex with a minor. According to the sheriff's department, before the suspects arrived at the rendezvous, most of them had already sent sexually explicit images to the "children" they were chatting with [128]. A similar sting operation in Lincolnton, North Carolina, in May 2015 led to 17 arrests [129]. Another sting operation in Galveston, Texas, in June 2015 led to the arrest of 11 men and women hoping to have sex with children [130].

In 2014 WTSP-TV aired an exposé critical of the sting operations in Polk County, Florida, alleging that detectives were resorting to more extreme methods to keep up their arrest rates. For example, an officer responds to an ad on an

adult dating site, pretending to be an adult woman, but once an online relationship begins to develop, the detective suggests the man might be interested in having sex with “her” child. Some police tactics have resulted in cases being thrown out of court, with judges making comments such as these:

- “It was the agent who repeatedly steered the conversation back to sexual activity with a minor.”
- “The government made a concerted effort to lure him into committing a crime.”
- “The law does not tolerate government action to provoke a law-abiding citizen to commit a crime” [131].

3.10.4 Ethical Evaluations of Police Sting Operations

Is it morally right for police detectives to pose as children on the Internet to entrap people who seek to have sex with children?

Utilitarian Analysis

Let's consider the various consequences of such a sting operation. A person allegedly interested in having sex with an underage minor is arrested and charged with attempted child rape. Suppose the person is found guilty and must serve time in prison. The direct effects of the sting operation are the denial of one person's freedom (a harm) and an increase in public safety (a benefit). Since the entire public is safer and only a single person is harmed, this is a net good.

The sting operation also has indirect effects. Publicity about the sting operation may deter other people who seek sex with children. This, too, is a beneficial result. It is harder to gauge how knowledge of sting operations influences innocent citizens. First, it may reduce citizens' trust in the police. Many people believe that if they are doing nothing wrong, they have nothing to fear. Others may become less inclined to provide information to the police when requested. Second, sting operations can affect everyone's online experiences. They demonstrate that people are not always who they claim to be. This knowledge may make people less vulnerable to being taken advantage of, but it may also reduce the amount of trust people have in others. Sting operations prove that supposedly private conversations can actually be made public. If instant messaging conversations lack honesty and privacy, people will be less willing to engage in serious

conversations. As a result, instant messaging loses some of its utility as a communications system. How much weight you give to the various consequences of police sting operations using instant messaging determines whether the net consequences are positive or negative.

Kantian Analysis

A Kantian focuses on the will leading to the action rather than the results of the action. The police are responsible for maintaining public safety. People who engage in pedophilia endanger innocent children. Therefore, it is the duty of police to try to prevent them from accomplishing what they intend to do. The will of the police detective is arrest people who engage in pedophilia. This seems straightforward enough.

If we dig a level deeper, however, we run into trouble. In order to arrest a person who wishes to engage in pedophilia, the police must identify this person. Since a person seeking sex with a child unlikely to confess on the spot if asked a question by a police officer, the police lay a trap. In other words, the will of the police detective is to deceive a person who engages in pedophilia in order to catch them. To a Kantian, lying is wrong, no matter how noble the objective. By collecting evidence of instant-messaging conversations, the police detective also violates the presumed privacy of this

medium. While police officers have a duty to protect the public safety, it is wrong for them to break other moral laws in order to accomplish this purpose. From a Kantian point of view, the sting operation is morally wrong.

Social-Contract-Theory Analysis

An adherent of social contract theory could argue that in order to benefit everyone, there are certain moral rules that people communicating online ought to follow. For example, people ought to be honest, and conversations ought to be kept confidential. By misrepresenting identity and/or intentions, the person engaging in pedophilia has broken a moral rule and ought to be punished. In conducting sting operations, however, police detectives also misrepresent their identities and record everything typed by suspects. The upholders of the law have broken the rules, too.

Furthermore, we have the presumption of innocence until proof of guilt. What if the police detective, through miscommunication or bad judgment, actually entraps someone who is not a person who engages in pedophilia? In this case, an innocent person has not broken any rules. They were simply in the wrong place at the wrong time. Yet society, represented by the police detective, did not provide the benefits instant messaging users expect to receive (honest

communications and privacy). In short, there is a conflict between society's need to punish a wrongdoer and its expectation that everyone (including the agents of the government) abide by its moral rules.

Summary of Ethical Analyses

To summarize our ethical evaluation of police sting operations, the actions of the police seem immoral from a Kantian point of view. Evaluations using the other ethical theories do not yield a clear-cut endorsement or condemnation of the stings. While the goals of the police are laudable, they accomplish their goals through deception and by revealing details of conversations thought to be private. Sting operations are more likely to be viewed as morally acceptable by someone who is more focused on the results of an action than the methods used; in other words, a consequentialist.

3.10.5 False Information

The Web is a more open communication medium than newspapers, radio stations, or television stations. Individuals or groups whose points of view might never be published in a newspaper or broadcast on a television or radio show may create an attractive Web site. The ease with which people

may get information out via the Web is one of the reasons the Web contains billions of pages. However, the fact that no one has to review a Web page before it is published means the quality of information available on the Web varies widely.

You can find many Web sites devoted to the American manned space program. You can also find many Web sites that argue the Moon landings were a hoax by NASA. Many Web sites describe the Holocaust committed by the Nazis before and during World War II. Other sites falsely claim the Holocaust never happened. A TikTok post claiming that the Titanic never sank attracted more than 11 million views before it was deleted [132].

Disputes about commonly held assumptions did not begin with the Web. Some television networks and newspapers are well known for giving a forum to people who question information provided through government agencies. Twice in 2001, the Fox TV network aired a program called “Conspiracy Theory: Did We Land on the Moon?” The program concludes NASA faked the Moon landing in the Nevada desert. Supermarket tabloids are notorious for their provocative, misleading headlines. Experienced consumers take into account the source of the information.

Most people would agree that *60 Minutes* on CBS is a more reliable source of information than the *Conspiracy Theory* series. Similarly, people expect information they find in the *New York Times* to be more reliable than the stories they read in a tabloid.

In traditional publishing, various mechanisms are put in place to improve the quality of the final product. For example, before Addison-Wesley published the first edition of this book, an editor sent draft copies of the manuscript to a dozen reviewers who checked it for errors, omissions, or misleading statements. I revised the manuscript to respond to the reviewers' suggestions. After I submitted a revised manuscript, a copy editor made final changes to improve the readability of the text, and a proofreader corrected typographical errors. Every subsequent edition has benefited from the feedback of experienced and knowledgeable college professors reviewing the proposed changes.

Web pages, on the other hand, can be published without any review. As you're undoubtedly well aware, the quality of Web pages varies dramatically. Fortunately, search engines can help people identify those Web pages that are most relevant and of the highest quality. Let's take a look at how the Google search engine does this.

The Google search engine keeps a database of many billions of Web pages. A software algorithm called PageRank ranks the quality of these pages. PageRank invokes a kind of voting mechanism. If Web page A links to Web page B, then page B gets a vote. However, all votes do not have the same weight. If Web page A is itself getting a lot of votes, then page A's link to page B gives its vote more weight than a link to B from an unpopular page.

When a user makes a query to Google, the search engine first finds the pages that closely match the query. It then considers their quality (as measured by the voting algorithm) to determine how to rank the relevant pages.

3.10.6 Cyberbullying

In November 2002, Ghyslain Raza, a high-school student living in Quebec, Canada, borrowed a videotape and used one of the high school's video cameras to film himself awkwardly swinging a golf ball retriever like a light saber, à la Darth Maul in *Star Wars Episode I*. A few months later, the owner of the videotape discovered the content and shared it with some friends. After one of them digitized the scene and made it available on the Internet, millions of people downloaded the file in the first two weeks [133]. Ghyslain

was nicknamed “the Star Wars kid,” endured prolonged harassment from other students, and eventually dropped out of school [134]. By 2006 the video had been viewed more than 900 million times [135].

Cyberbullying is the use of the Internet or the phone system to inflict psychological harm on another person. Frequently, a group gangs up to cyberbully the victim. Examples of cyberbullying include the following:

- Repeatedly texting or emailing hurtful messages to another person
- Spreading lies about another person
- Tricking someone into revealing highly personal information
- “Outing” or revealing someone’s secrets online
- Posting embarrassing photographs or videos of other people without their consent
- Impersonating someone else online in order to damage that person’s reputation
- Threatening or creating significant fear in another person

Surveys have revealed that cyberbullying is common among teenagers. Cox Communications surveyed 655 American teenagers in 2009, and 19 percent reported that they had been cyberbullied online, via cell phone, or through both media. Ten percent of the teenagers admitted to cyberbullying someone else. When asked why they had cyberbullied someone else, the most common responses were “they deserved it” and “to get back at someone” [110].

In some instances cyberbullying has led to the suicide of the victim, as in the case of 13-year-old Megan Meier. According to her mother, “Megan had a lifelong struggle with weight and self-esteem” [136]. She had talked about suicide in third grade, and ever since then she had been seeing a therapist [136]. Megan’s spirits soared when she met a 16-year-old boy named Josh Evans on MySpace. They flirted online for four weeks but never met in person. Then Josh seemed to sour on their relationship. One day he let her know that he didn’t know if he wanted to be friends with her anymore. The next day he posted the following [136, 137]:

You are a bad person and everybody hates you.

Have a shitty rest of your life.

The world would be a better place without you.

When Megan angrily responded to this post, others ganged up on her: “Megan Meier is a slut”; “Megan Meier is fat” [136]. Later that afternoon, Megan died by suicide in her bedroom.

Eventually, the community learned that “Josh Evans” did not exist. The MySpace account had been created just a couple of houses away from the Meier home by 18-year-old Ashley Grills, 13-year-old Sarah Drew, and Lori Drew, Sarah’s mother. Sarah had a falling out with Megan, and Ashley suggested creating the MySpace account to find out what Megan might be saying about Sarah. Lori Drew had approved the plan. Most of the messages from “Josh” had been written by Sarah or Ashley, but Lori Drew had been aware of what they were doing [138].

The county’s district attorney declined to prosecute Lori Drew because there was no Missouri law against cyberbullying [139]. The FBI investigated the case, however, and in 2008 federal prosecutors charged Drew with four felony counts under the Computer Fraud and Abuse Act for violating the MySpace terms of service. A jury found her not guilty of these crimes but did convict her of three misdemeanors [140]. In 2009 a US district judge overturned these convictions, stating that criminal charges should not

have been brought against Drew for breaking a contract with an Internet service provider [141].

In April 2009, the Megan Meier Cyberbullying Prevention Act was introduced in the US House of Representatives. The purpose of the proposed law was to “impose criminal penalties on anyone who transmits in interstate or foreign commerce a communication intended to coerce, intimidate, harass, or cause substantial emotional distress to another person, using electronic means to support severe, repeated, and hostile behavior” [142]. Some civil libertarians objected to the proposed legislation, arguing that it would take away free-speech rights guaranteed under the First Amendment to the US Constitution. The law did not win approval by the House of Representatives.

3.10.7 Revenge Porn

The *Collins English Dictionary* defines **revenge porn** to be “a pornographic image or film which is published, posted (e.g., on the Internet), or otherwise circulated without the consent of one or more of the participants, usually with malicious and vindictive intent, such as following a break-up” [143]. Revenge porn is a special case of cyberbullying.

The Web site Is Anyone Up?, launched in 2010 by Hunter Moore, quickly became a magnet for revenge porn submissions, significantly raising the visibility of this issue. Moore appeared on the talk show *Anderson*, along with two women whose images had appeared on the site without their permission. When asked by Anderson Cooper whether he felt bad about what he had done “to these two women,” Moore replied to the women, “No one . . . made you take these pictures. It’s 2011; everything’s on the Internet” [144]. One of the women responded, “If we’re choosing to send them to someone we trust, that doesn’t mean we expect them to show up on the Internet” [144]. Moore shut down the Web site in April 2012. In February 2015 he pleaded guilty to one count of accessing a protected computer to obtain information for financial gain and one count of aggravated identity theft, and in December 2015 he was sentenced to two-and-a-half years in federal prison [145].

Posting revenge porn has been criminalized in Australia, France, Germany, Israel, Italy, Japan, Malta, the Philippines, Singapore, South Africa, South Korea, the United Kingdom, and the United States. In 2011, after Noe Iniguez broke up with his girlfriend, he posted topless photos of her on her employer’s Facebook page. Three years later Iniguez became

the first person to be convicted under California's law criminalizing the unauthorized posting of nude images [146].

In the first six months of 2015, three corporations took action to protect victims of revenge porn. Reddit announced it was banning the posting of sexually explicit images without the consent of the people in the images [147]. Twitter updated its rules to include, "You may not post intimate photos or videos that were taken or distributed without the subject's consent," and it promised to remove links to such content [148]. Google announced that it would create an online form enabling victims to request that nude or sexually explicit images of themselves posted without their consent be withheld from Google search results [149].

3.11 Internet Addiction

Many people spend a great deal of time staring at screens. You probably know someone who is constantly texting or instant messaging. Perhaps you have a friend who plays computer games for hours on end. Why do people get so attached to their devices? Is it possible to become addicted to these activities?

According to psychiatrist Edward Hallowell, the brains of heavy users of digital devices crave newness, and every text or email message results in a release of the neurotransmitter dopamine that feeds the craving [150]. That's why we find it difficult to break away from these activities [151].

Another reason people can find it hard to leave their devices is because businesses are trying to keep them attached. The term **attention economy** refers to the idea that the attention of a consumer has economic value. Businesses vie for the attention of consumers through Web sites, social media, and other channels. They use technology to deliver personalized content designed to keep consumers engaged.

3.1 1.1 Is Internet Addiction Real?

Our electronic devices are like addictive drugs in the sense that when we use them, they provide our brains with immediate positive feedback [152]. Psychiatrist Jerald Block maintains that Internet addiction is a common disorder with at least three variants: gaming, “sexual preoccupations,” and messaging [153]. According to Block, all of these variants share the following characteristics: engaging in excessively long sessions online; becoming angry, tense, or depressed when forced to be away from a device; feeling a need to enhance online experiences by obtaining better equipment or spending more time online; and suffering negative social consequences.

Citing insufficient research data, the American Psychiatric Association did not include Internet addiction in its most recent *Diagnostic and Statistical Manual of Mental Disorders* [154]. On the other hand, Internet addiction has been recognized and taken seriously in both South Korea and China since 2007.

The average South Korean high-school student spends 23 hours per week gaming [155]. After 10 cardiopulmonary-related deaths in Internet cafés ([Figure 3.2](#)), South Korea began training mental health counselors in how to treat Internet addiction [156, 157]. In 2011 the South Korean government passed a law prohibiting children under the age of 16 from accessing online games between midnight and 6 a.m. [158].

Figure 3.2



Many South Koreans play online games in centers called PC bangs. In 2005 a 28-year-old South Korean man died after playing one game practically nonstop for 50 hours.

(Kim-Jae Hwan/AFP/Getty Images)

In 2007 the Chinese government issued a regulation limiting people under 18 to three hours of online game play per day, and in 2021 it put in place even stricter limits. Children and teenagers in China are no longer allowed to play online games on school days, and their weekend and holiday game play is limited to one hour per day [159].

Our concern in this section is excessive use of digital devices that causes harm. We will use the term “Internet addiction” to describe this behavior because it is the term most widely used by the press, even as we recognize that experts disagree about whether excessive use is a genuine addiction.

3.11.2 Contributing Factors to Addiction

According to Stanton Peele, social, situational, and individual factors can increase a person’s susceptibility to addiction. For example, peer groups play an important role in determining how individuals use alcohol and other drugs. People in stressful situations are more likely to become addicted, as are those who lack social support and intimacy, and those who have limited opportunities for “rewarding, productive activity” [160]. Individual factors that make a person more

susceptible to addiction include a tendency to pursue an activity to excess, a lack of achievement, a fear of failure, and feelings of alienation.

Kimberly Young's studies led her to "believe that behaviors related to the Internet have the same ability to provide emotional relief, mental escape, and ways to avoid problems as do alcohol, drugs, food, or gambling" [161]. She noted that those who are addicted to the Internet are typically addicted to a single application.

3.11.3 Ethical Evaluation of Internet Addiction

People who use digital devices excessively can harm themselves and others for whom they are responsible. For this reason, excessive use of digital devices is a moral issue.

Kantianism, utilitarianism, and social contract theory all share the Enlightenment view that individuals, as rational beings, have the capacity and the obligation to use their critical judgment to govern their lives [162]. Kant held that addiction is a vice, because it's wrong to allow your bodily desires to dominate your mind [163]. Mill maintained that some pleasures are more valuable than others and that people

have the obligation to help each other distinguish better pleasures from worse ones [96]. Ultimately, people are responsible for the choices they make. Even if an addict is “hooked,” the addict is responsible for choosing to engage in the activity the first time.

The Enlightenment view assumes that people are capable of controlling their compulsions. According to Jeffrey Reiman, vices are “dispositions that undermine the sovereignty of practical reason. Dispositions, like habits, are hard but not impossible to overcome, and undermining something weakens it without necessarily destroying it entirely” [162, p. 89].

Reiman’s view is supported by Peele, who believes addicts can choose to recover from their addictions. “People recover to the extent that they (1) believe an addiction is hurting them and wish to overcome it, (2) feel enough efficacy to manage their withdrawal and life without the addiction, and (3) find sufficient alternative rewards to make life without the addiction worthwhile” [160, p. 156].

While our analysis to this point has concluded that individual addicts are morally responsible for their addictions, it’s also possible for a society to bear some collective moral responsibility for the addictions of some of its members. We

have already discussed how social conditions can increase a person's susceptibility to addiction, and Peele states an addict will not recover unless life without the addiction has sufficient rewards. Some people use digital devices as a way to escape into their own world, because in the "real world" they suffer from social isolation [164]. Perhaps we should reflect on whether any of our actions or inactions make certain members of our community feel excluded.

Summary

The Internet and cellular networks are powerful and flexible technologies that have revolutionized communication by making it possible for billions of people to communicate with each other. Never before have so many people around the globe been able to interact: sharing information, engaging in commercial transactions, organizing political movements, competing in online games, helping those in need, and much more. At the same time, those who seek to exploit others have found new ways to do so using these same technologies.

The Web provides a remarkably simple way for people to post and access information, and it now contains several billion indexed pages. The Web has replaced printed encyclopedias and is giving books, journals, and newspapers a run for their money. It contains images of sublime beauty and shocking cruelty, uplifting poetry and expletive-ridden hate speech, well-organized encyclopedias and figments of paranoid imaginations, useful software and hidden computer viruses. In short, it is a reflection of the best and the worst of humanity.

Texting has supported myriad activities, from the social to the deadly serious. It allows friends and family members to check in on each other and set up social encounters, it is an important way for people to make international money transfers without banks, and it has also been used to organize mass protests and overthrow governments.

As social media have become more popular, they have become an increasingly important way for people to get the news. In fact, two-thirds of Americans now read news stories on social media sites, principally Facebook. Facebook's algorithms help ensure people will see news stories that reinforce their views rather than challenge them. For this reason, some commentators fear that Facebook may be harming democracy by increasing political polarization.

Governments have responded to the latent power of these new technologies in a variety of ways. The most repressive governments have made the Internet inaccessible to the masses. Other governments have instituted controls that prevent certain sites from being accessed. To promote social and political stability, the Chinese government prohibits access to certain Internet sites and sometimes restricts cell phone service. On one occasion it shut down the Internet in an entire region for months. In an effort to ensure children

are not exposed to online pornography, the United States government requires libraries to install antipornography filters. Given the focus in the United States on preventing children from being exposed to pornography, it is ironic that many minors have voluntarily created and transmitted explicit images of themselves.

Those who wish to exploit other people have found ways to use these new communication technologies to achieve their ends. Spammers are sending out hundreds of billions of spam emails every day. Others make a living selling stolen credit card information. Millions of people every year are the targets of identity theft.

The challenge for each of us is to reflect on our own actions and ponder whether there are times we may have stepped over the line and put ourselves first, privileging our own needs and desires over those of other people. Cyberbullying and posting revenge porn are the result of decisions made by individuals.

Finally, these new communication technologies have provided new opportunities for people to harm themselves. A good example is the phenomenon commonly referred to as Internet addiction. Some people become so engaged with their digital devices that they spend too much time with their

eyes glued to their screens and too little time taking care of their responsibilities to themselves and to others.

What is the proper governmental response to the Internet addiction crisis? The governments of South Korea and China have put limits on the number of hours per day that children can play online games, and an educational campaign in South Korean schools informs children of the dangers of Internet addiction and encourages healthy alternative activities, such as listening to music.